

ArenaMind

PRD(Product Requirement Document)

Part 2:

Product Goals & Objectives

ArenaMind AI aims to become the intelligence layer connecting every operational component inside a FIFA World Cup stadium.

Unlike existing solutions that address isolated operational problems, ArenaMind AI continuously observes, reasons, predicts, and recommends actions across the entire tournament ecosystem.

Objectives

Primary Objective

Build a **Generative AI Operating System** capable of orchestrating real-time stadium operations while improving the experience for:

-  Fans
-  Stadium Operators
-  Security Teams
-  Volunteers
-  Medical Teams
-  Transportation Authorities
-  Accessibility Services
-  Sustainability Teams

through a single, unified AI platform.

Business Objectives

Objective	Description
Improve Fan Experience	Reduce navigation friction and waiting times
Improve Operational Efficiency	Centralize operational intelligence
Improve Decision Making	Shift from reactive workflows to predictive recommendations
Improve Safety	Detect and mitigate operational risks before they escalate
Improve Sustainability	Optimize energy, waste, and transportation resources
Improve Accessibility	Deliver inclusive experiences for all attendees

Technical Objectives

ArenaMind AI should:

- Process live operational events
 - Support natural language interaction
 - Generate contextual recommendations
 - Simulate operational scenarios
 - Integrate with existing stadium systems
 - Scale across multiple venues simultaneously
 - Maintain low-latency responses
 - Continue operating even if one LLM provider becomes unavailable
-

Success Metrics

ArenaMind AI's success will be measured using quantitative and qualitative metrics.

Fan Experience KPIs

KPI	Target
Navigation Time Reduction	30%
Queue Waiting Time Reduction	25%
AI Assistant Satisfaction	>90%
Route Accuracy	>95%
Fan Engagement	>80%

Operational KPIs

KPI	Target
Incident Detection Time	<15 seconds
AI Recommendation Accuracy	>90%
Decision Support Response Time	<5 seconds
Operational Dashboard Uptime	99.9%
Prediction Confidence	>85%

Sustainability KPIs

KPI	Target
Energy Optimization	15%
Waste Reduction	20%
Carbon Footprint Reduction	10%
Resource Utilization	+25%

AI KPIs

KPI	Target
Average Response Time	<3 seconds
Hallucination Rate	<3%

KPI	Target
Translation Accuracy	>95%
Vision Detection Confidence	>90%
AI Availability	99.95%

Product Principles

ArenaMind AI follows five core product principles.

1. AI First

Artificial intelligence is the primary interface—not an optional feature.

Every workflow should leverage AI to simplify complex operational tasks.

2. Human-in-the-Loop

ArenaMind AI recommends actions.

Humans remain responsible for final operational decisions.

Critical actions require operator confirmation.

3. Real-Time by Default

Every dashboard, prediction, and recommendation should reflect live operational conditions.

Static reports are secondary.

4. Accessibility by Design

Every feature must support:

- Screen Readers
- Voice Navigation
- Keyboard Navigation

- Color Blind Modes
- High Contrast Themes
- Large Text

Accessibility is a core requirement—not an enhancement.

5. Explainable AI

Every recommendation should answer:

- Why?
- Based on what data?
- How confident is the prediction?
- What happens if ignored?

Example:

| Crowd congestion predicted near Gate C in 12 minutes.

Reason:

- Two shuttle buses arriving
- Food court demand increasing
- Gate occupancy already at 82%

Confidence:

94%

Recommended Action:

Open Gate D and redirect incoming visitors.

User Personas

ArenaMind AI serves multiple stakeholders across the tournament ecosystem.

Persona 1 — Stadium Operations Manager

Goals

- Maintain smooth stadium operations
- Prevent operational disruptions
- Monitor every venue in real time

Pain Points

- Too many disconnected dashboards
- Delayed incident awareness
- Limited predictive insights

ArenaMind Helps By

- AI-generated operational summaries
 - Predictive alerts
 - Natural language operational queries
 - A unified command center
-

Persona 2 — Security Officer

Goals

- Maintain public safety
- Prevent unauthorized access
- Respond quickly

Pain Points

- CCTV overload
- Limited contextual awareness

- Slow communication

ArenaMind Helps By

- AI incident summaries
 - Crowd anomaly detection
 - Emergency simulations
 - Intelligent routing
-

Persona 3 — Volunteer

Goals

Help visitors quickly.

Pain Points

- Manual task allocation
- Communication delays
- Frequent operational changes

ArenaMind Helps By

- AI task assignment
 - Voice assistant
 - Multilingual translation
 - Smart navigation
-

Persona 4 — Fan

Goals

Enjoy the match.

Pain Points

- Long queues
- Transportation confusion
- Language barriers
- Finding facilities

ArenaMind Helps By

- AI itinerary
 - Smart navigation
 - Queue prediction
 - Live transportation
 - Food recommendations
-

Persona 5 — Accessibility User

Goals

Navigate independently.

Pain Points

- Limited accessible information
- Difficult navigation
- Emergency uncertainty

ArenaMind Helps By

- Voice-first interaction
- Wheelchair routes
- Indoor navigation
- Audio guidance

- Live accessibility support

Persona 6 — Sustainability Manager

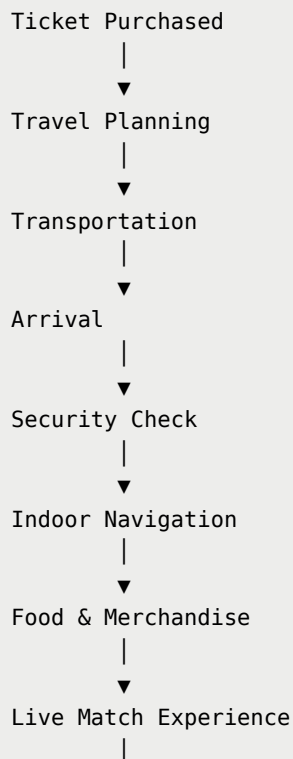
Goals

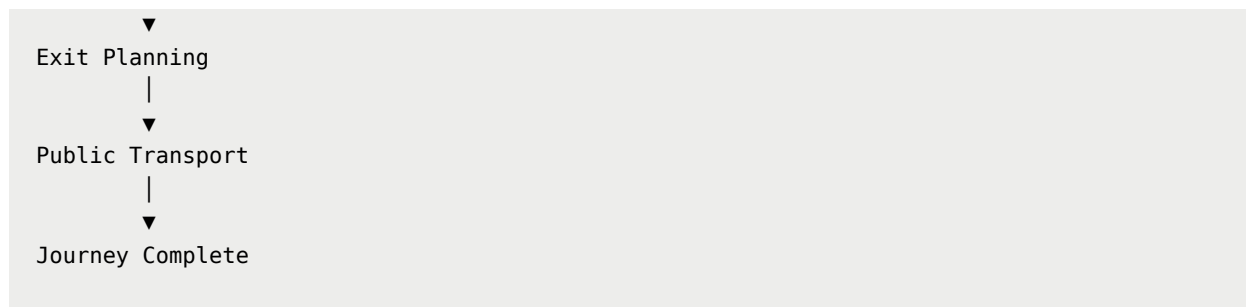
Reduce environmental impact.

ArenaMind Helps By

- Carbon analytics
- Waste prediction
- Energy optimization
- AI sustainability recommendations

User Journey Overview





ArenaMind AI supports users throughout the complete journey.

Product Modules

ArenaMind AI consists of eleven integrated intelligence modules.

Module	Description
 AI Command Center	Central operational intelligence
 Crowd Prediction Engine	Predict congestion before it occurs
 Smart Fan Assistant	Personalized navigation & recommendations
 Transport Intelligence	Live transport optimization
 Accessibility Copilot	Inclusive navigation & assistance
 Sustainability Optimizer	Energy & waste optimization
 Emergency Response Copilot	AI-assisted incident response
 Volunteer Intelligence	Dynamic workforce management
 Stadium Digital Twin	Interactive operational simulation
 AI Media Generator	Multilingual announcements & updates
 Executive Analytics	Post-match insights & reporting

MVP Scope

The hackathon MVP focuses on demonstrating the platform's core intelligence capabilities.

Included

- AI Command Center
 - Fan Assistant
 - Digital Twin Visualization
 - Crowd Heatmap
 - AI Incident Summaries
 - Natural Language Queries
 - Gemini + Groq AI Orchestration
 - Interactive Stadium Map
 - Transportation Dashboard
 - Executive Analytics
-

Excluded

The following are planned for future releases:

- Live CCTV Integration
- Production IoT Infrastructure
- Real Payment Systems
- Ticket Purchasing
- Identity Verification
- Wearable Device Integration
- Drone Control
- Smart Parking Hardware

These features require production infrastructure and are outside the scope of a five-day hackathon.

Feature Prioritization

Priority	Feature
★★★★★	AI Command Center
★★★★★	Fan Assistant
★★★★★	Crowd Prediction
★★★★★	Digital Twin
★★★★★	Emergency Response
★★★★★	Volunteer Intelligence
★★★★★	Transport Intelligence
★★★★	Sustainability Dashboard
★★★★	Accessibility Copilot
★★★★	Executive Reports
★★★	AI Media Generator

Definition of Success

ArenaMind AI succeeds if judges can experience a realistic scenario in which the platform:

- predicts congestion before it forms,
- answers operational questions using natural language,
- visualizes a stadium digital twin,
- provides multilingual assistance,
- recommends mitigation actions,
- and demonstrates seamless AI orchestration across multiple operational domains.

The prototype should leave judges with the impression that they are interacting with the intelligent operating system of a future FIFA World Cup stadium—not simply another AI chatbot or analytics dashboard.